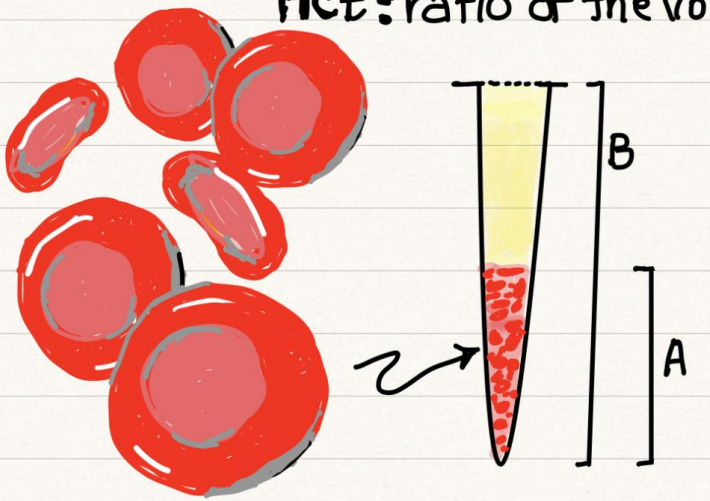
**Hct:** ratio of the vol. of RBCs to vol. of whole blood.

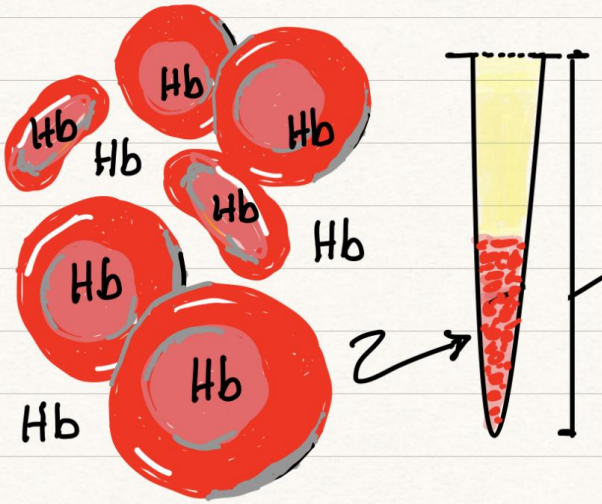



$$Hct = \frac{A\ \text{vol.}}{B\ \text{vol.}} = \frac{L\ \text{RBCs}}{L\ \text{blood}}$$

or %

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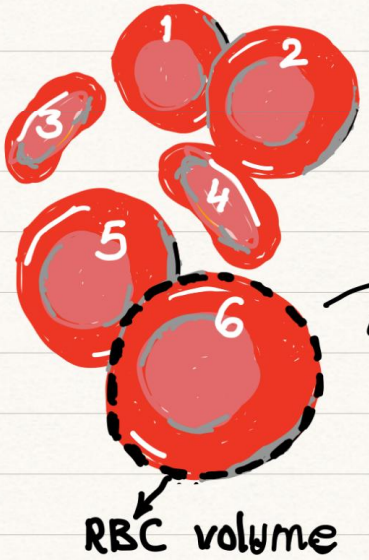
**Hb conc.:** Amount of Hb in vol. of blood.




$$C = \frac{M}{V}$$

$$\text{Hb conc.} = \frac{\text{Mass}}{\text{vol.}} = \frac{g}{dL}$$

## MCV: Volume of one RBC



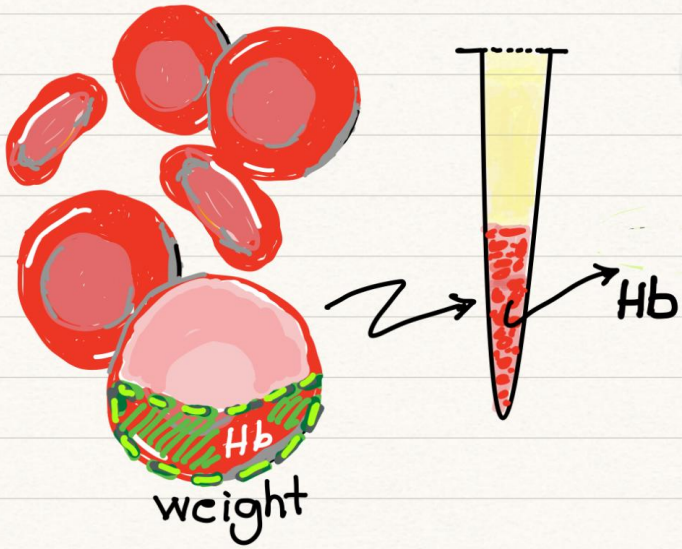

volume of all RBCs in the Sample ≡ Hct

$$MCV = \frac{Hct}{\text{RBCs count}} = \frac{L/L}{\#/L}$$

volume units "fL" $= 10^{-15}$ L

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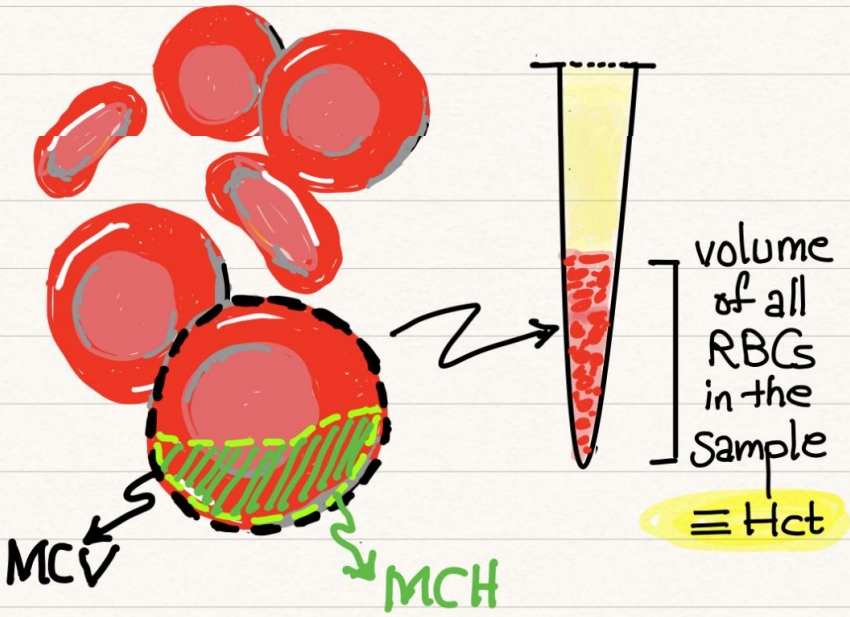
## MCH: Amount of Hb in one RBC




$$MCH = \frac{Hb\ Conc.}{\text{RBCs count}} = \frac{g/dL}{\#/L}$$

Mass unit "pg" $= 10^{-12}$ g

**MCHC :** Proportion of each RBC taken up by Hb.




$$MCHC = \frac{Hb.conc}{Hct} = \frac{g/dL}{L/L} = g/dL$$

$$MCHC = \frac{MCH}{MCV} \cong 30\% \text{ of a RBCs is Hb}$$